Big Data Project Proposal

Scalable Analysis and Demand Prediction of New York City Taxi Trip Data Using Apache Spark

Course: Big Data Analytics

Section: [C]

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Project Abstract

New York City taxi trip records represent a large-scale real-world dataset that captures urban mobility patterns through millions of trips recorded every month. Each trip contains useful details such as pickup and drop-off timestamps, trip distance, fare amount, passenger count, and pickup/drop-off location identifiers. Because of the size and complexity of this dataset, it is well suited for distributed processing and analysis using Big Data tools.

In this project, we propose to use Apache Spark and PySpark to build a scalable analytics pipeline for New York City Yellow Taxi trip data. The project will include data ingestion, preprocessing, distributed analysis, and visualization of key travel patterns such as peak demand hours, busiest taxi zones, trip duration trends, and fare behavior. In addition to descriptive analytics, we also plan to include a predictive component to estimate demand at different times and locations, and an anomaly detection component to identify unusual trip or fare patterns.

The overall goal of this project is to demonstrate how distributed data processing frameworks can be used to efficiently analyze millions of transportation records and generate meaningful insights from large urban mobility datasets.

Keywords: Big Data, Apache Spark, PySpark, Spark SQL, Urban Mobility, Demand Prediction, Anomaly Detection

1 Problem Statement

Modern cities generate extremely large volumes of transportation data every day. While this data contains useful information about travel demand, traffic behavior, trip efficiency, and fare trends, extracting insights from it is not straightforward using traditional single-machine processing tools. As the dataset grows into tens of millions of records, tasks such as cleaning, aggregation, and pattern analysis become computationally expensive and slow.

The New York City Yellow Taxi Trip dataset is a strong example of such large-scale urban data. Since it contains trip-level records over many months and years, it can be used to study mobility trends in a meaningful way. However, because the dataset is large and has multiple attributes, it requires a scalable distributed processing framework.

The main goal of this project is to use Apache Spark to process and analyze NYC taxi trip data efficiently. Beyond descriptive analysis, we also want to strengthen the project by developing a simple predictive model for taxi demand and by detecting anomalous trips such as unusually high fares, very long trip durations, or extreme trip distances.

2 Project Objectives

The objectives of this project are as follows:

- To collect and process large-scale NYC Yellow Taxi trip data using Apache Spark and PySpark.
- To design a complete Big Data pipeline covering data ingestion, cleaning, distributed transformation, analysis, and visualization.
- To preprocess the data by handling null values, removing invalid records, standardizing timestamps, and selecting relevant attributes.
- To perform descriptive analytics on the dataset, including:
  - busiest pickup and drop-off zones,
  - peak travel hours and days,
  - trip distance and fare distributions,
  - passenger count patterns,
  - average trip duration across locations.
- To implement a predictive analysis component for estimating taxi demand based on time and pickup zone features.
- To implement an anomaly detection component to identify suspicious or unusual trips, such as:
  - extremely high fares,
  - unusually long durations,
  - abnormal trip distances,

  - low-fare or zero-fare outliers.

- To visualize the final analytical results using Python-based plotting libraries.
- To demonstrate the usefulness of distributed computing for handling millions of real-world records efficiently.

3 Data Source Information

Dataset Name: NYC TLC Yellow Taxi Trip Records

Primary Data Source Page:

`https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page`

Specific Dataset Files Proposed for Initial Project Scope:

- January 2023 Yellow Taxi data:
  `https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2023-01.parquet`
- February 2023 Yellow Taxi data:
  `https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2023-02.parquet`
- March 2023 Yellow Taxi data:
  `https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2023-03.parquet`

These monthly parquet files are directly downloadable and provide a clearly verifiable dataset source for the project.

Dataset Structure:

Each record contains trip-related attributes such as:

- pickup and drop-off datetime,
- pickup and drop-off location IDs,

- passenger count,
- trip distance,
- fare amount,
- payment type,
- additional fare-related fields.

Data File Size:

The exact file sizes vary by month, but the combined data for multiple months is large enough for distributed analysis and typically falls within the range of several hundred megabytes to multiple gigabytes depending on the selected period.

Approximate Number of Records:

Each monthly Yellow Taxi dataset contains millions of trip records. Using three months of data is expected to produce a working dataset with tens of millions of rows, which is appropriate for a Big Data course project.

4 Proposed Project Pipeline

The planned pipeline for the project is as follows:

1. **Data Ingestion:** Download the selected Yellow Taxi parquet files and load them into the processing environment.

2. **Distributed Processing using Spark:** Read the parquet files into Spark DataFrames and perform scalable preprocessing such as filtering invalid rows, selecting relevant columns, and deriving new time-based features.

3. **Analytical Processing:** Use Spark SQL and PySpark transformations to compute trip statistics, demand patterns, zone-level aggregations, and time-based summaries.

4. **Predictive Analysis:** Build a simple demand prediction model using selected features such as hour of day, day of week, and pickup zone.

5. **Anomaly Detection:** Detect outlier trips using rule-based thresholds or simple statistical methods over fare, distance, and duration fields.
6. **Visualization and Reporting:** Export summarized results and create charts and plots in Python using Matplotlib and Seaborn for the final report.

This pipeline gives the project a more complete Big Data workflow rather than limiting it to only descriptive analytics.

5 Proposed Technologies and Programming Language

The technologies proposed for this project are listed below. These may be updated slightly during implementation depending on the execution environment and project progress.

Programming Language

- Python

Big Data Processing Framework

- Apache Spark
- PySpark
- Spark SQL

Data Storage / Input Format

- Parquet files

Data Analysis and Modeling

- PySpark DataFrames
- Spark SQL

- Pandas
- Scikit-learn (for baseline predictive modeling if needed)

Visualization

- Matplotlib
- Seaborn
- Jupyter Notebook

Execution Environment

- Dataproc Hadoop Environment / Google Cloud Platform

6 Expected Outcome

By the end of this project, we expect to build a scalable Spark-based analytics workflow that can process millions of taxi trip records efficiently. The project will generate useful insights into urban travel patterns in New York City while also including a stronger analytical scope through demand prediction and anomaly detection. This will make the project more technically complete and more aligned with the expectations of a Master's-level Big Data course.